

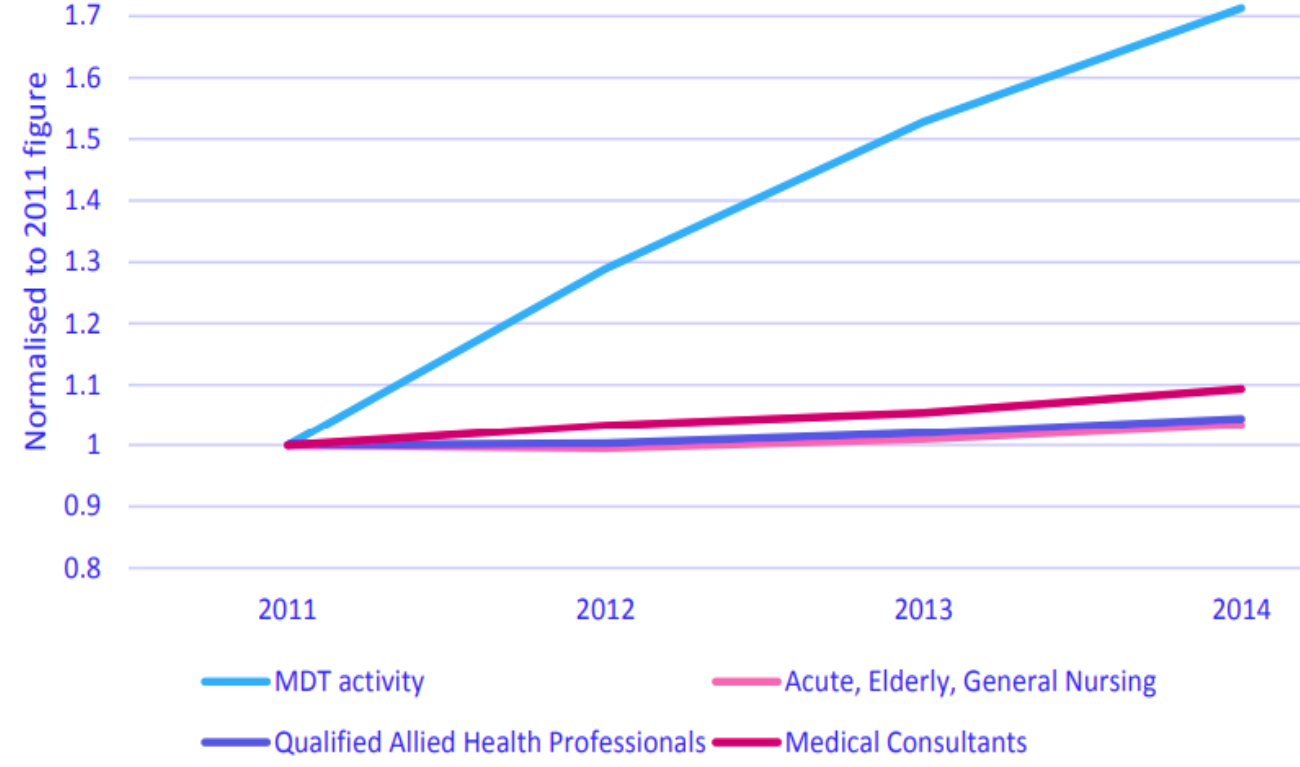
Lightweight Model Augmented By Expert Knowledge In Realistic Clinical Decision-Making On Colorectal Cancer Treatment

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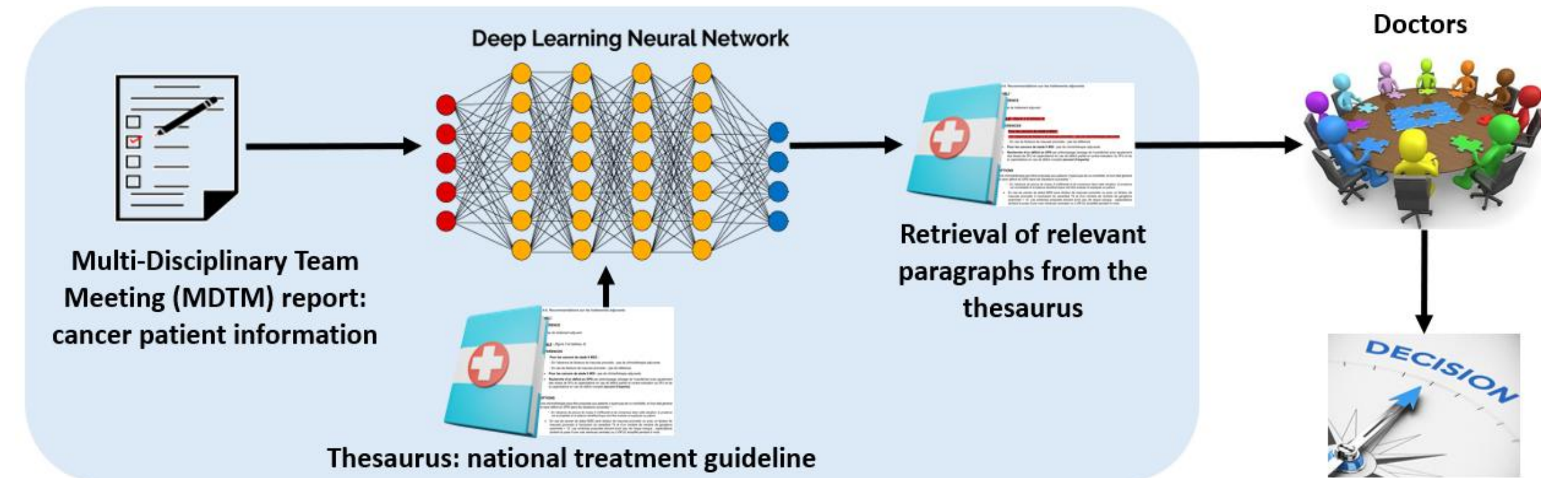
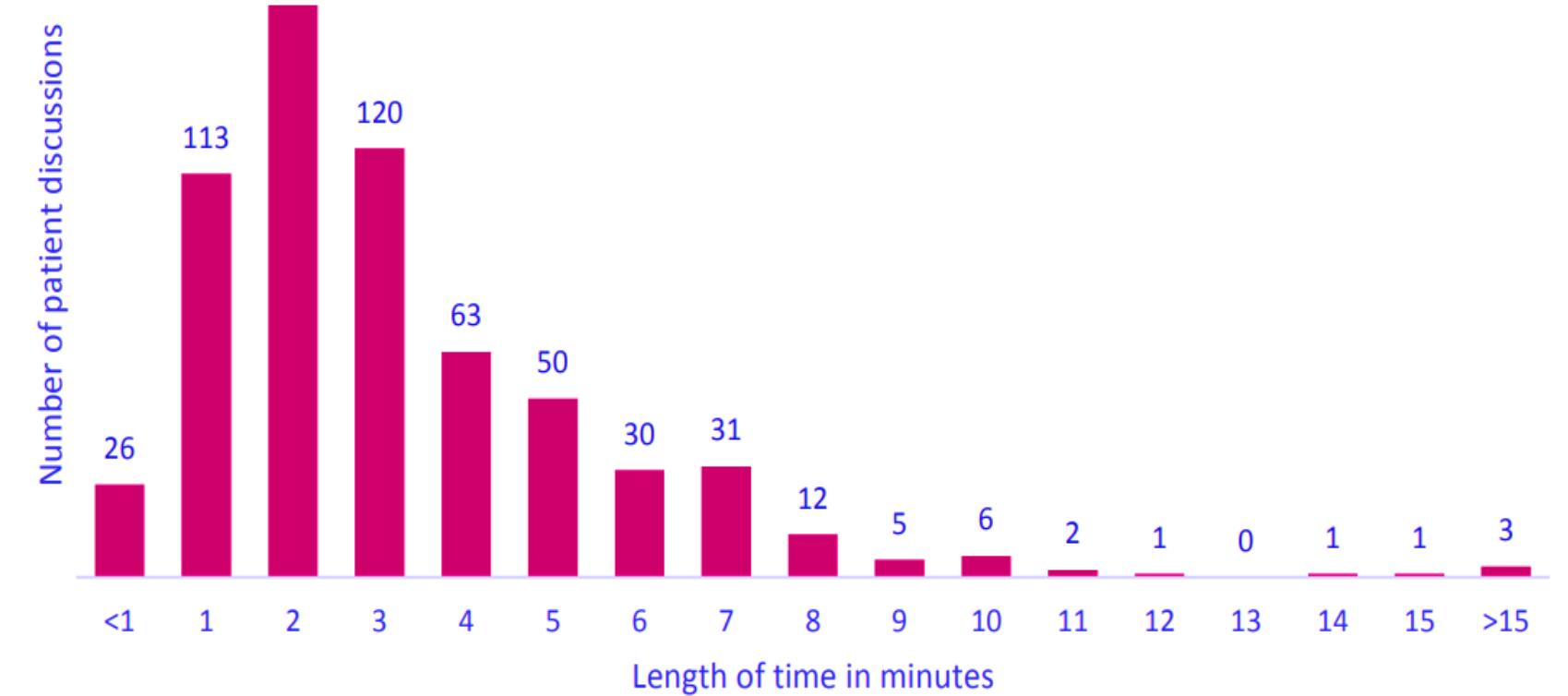
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BACKGROUND

"Sometimes we discuss up to 70 patients. This is after a whole day of clinics and we don't finish until gone 19.00. Would you want to be number 70?" (Upper GI MDT Coordinator)



"Current MDT can last 2+ hours. I still have no allowance in job plan for any of this and cannot stay for more than 1 hour of meeting due to clinical commitments. With half the substantive Radiology consultant staff in post at my hospital, the pressure is on direct clinical care, rather than MDT." (Lung Radiologist)



Goal:

Retrieving—for each patient—the relevant paragraphs of treatment recommendations from the thesaurus.

Challenges:

- Limited availability of annotated clinical data.
- Computational constraints: limited hospital resources.
- Patient data privacy: on-premises AI model requirement.

Background:

- **Colorectal cancer:** among the deadliest and most frequent cancers.
- **Thesaurus:** national recommendation guidelines detailing the best treatment for various patient situations.
- **Multidisciplinary Team Meeting (MDTM):** team of healthcare professionals discussing treatments for each patient.
- **Issues:** determining the best treatment is a complex task due to the growing number of patient records to review in a limited time, occasional unavailability of experts, and varying levels of familiarity with the latest guidelines.

DATASET

MTDM reports

STATUT THERAPEUTIQUE	
Statut Thérapeutique	Date Début Traitement :
Traitements antérieurs	
Commentaires Traitements antérieurs :	
CLASSIFICATION TNM	
TNM Clinique :	My :
Commentaires :	
AUTRES CLASSIFICATIONS	
Autre Classification :	
State :	
Commentaires :	
DESCRIPTIF HISTOLOGIQUE	
Date prélèvement :	Tumeur unique :
Code Adicap :	
HISTOLOGIE	
TNM Anapath :	My Perm :
Grade Score :	Residu tumoral :
Infos Spécifs :	

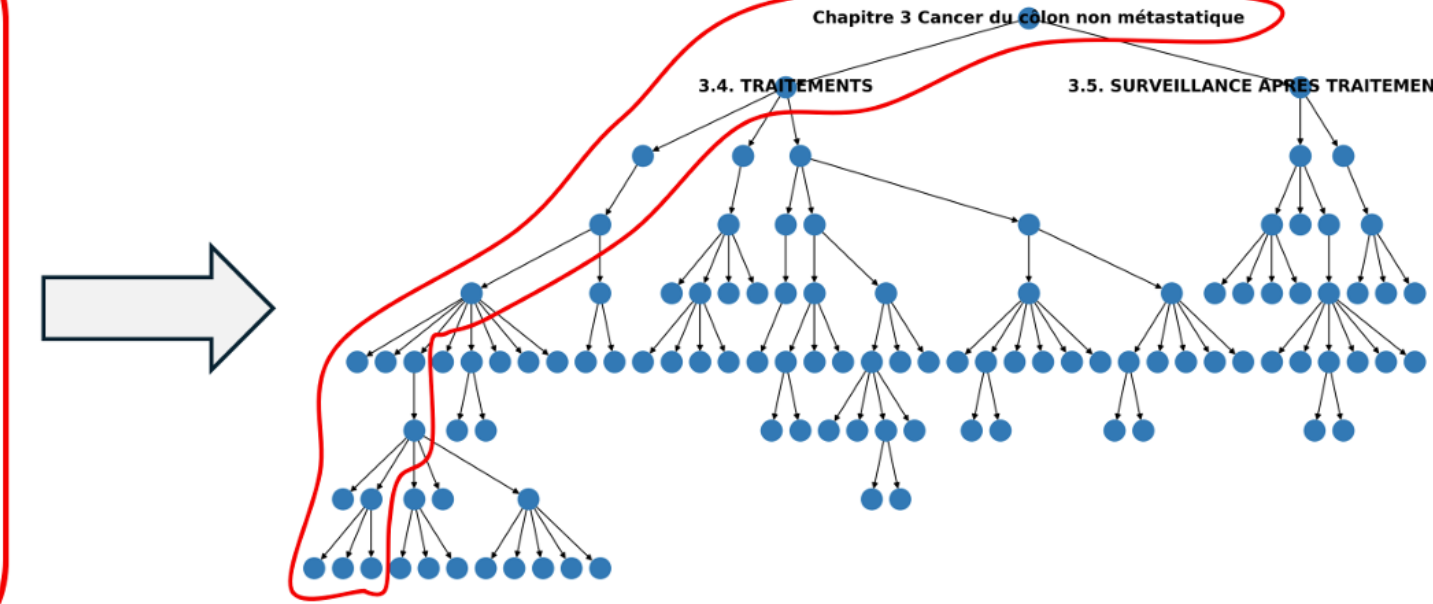
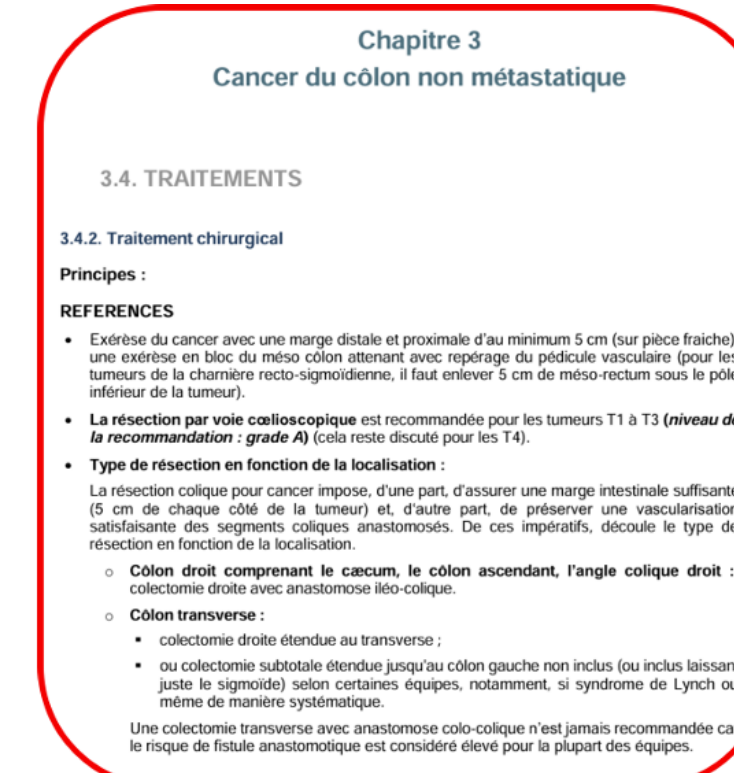
- 222 de-identified French medical MDTM reports from patients with non-metastatic colon cancer.

- Obtained from Nice University Hospital.

- MDTM reports follow a standardized template—a form with fillable fields—summarizing key patient details.

- MDTM reports are usually 2 pages long.

Non-metastatic colorectal cancer « thesaurus »



- The thesaurus is a long and highly structured document organized into multiple hierarchical levels of titles, sections, subsections, lists, paragraphs, etc.

Competing approaches : LLMs

LLM characteristics:

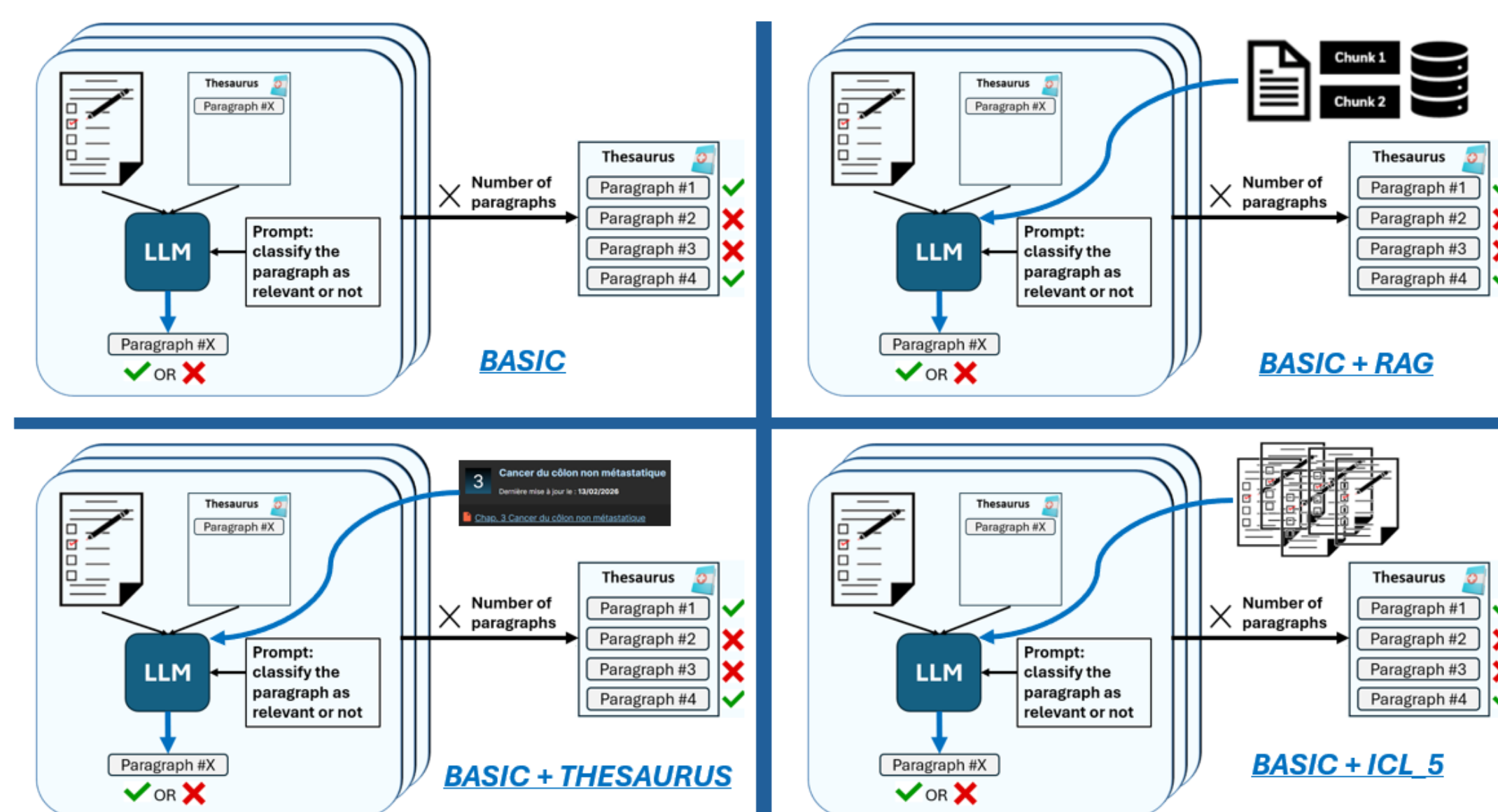
- Open-weight models
- Parameters: 7B to 47B
- Context window: 32k to 128k tokens
- Instruction-tuned versions
- Fine-tuned on medical or French data

Metric: F2-score

$$F_2 = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(\beta^2 \cdot \text{Precision}) + \text{Recall}}$$

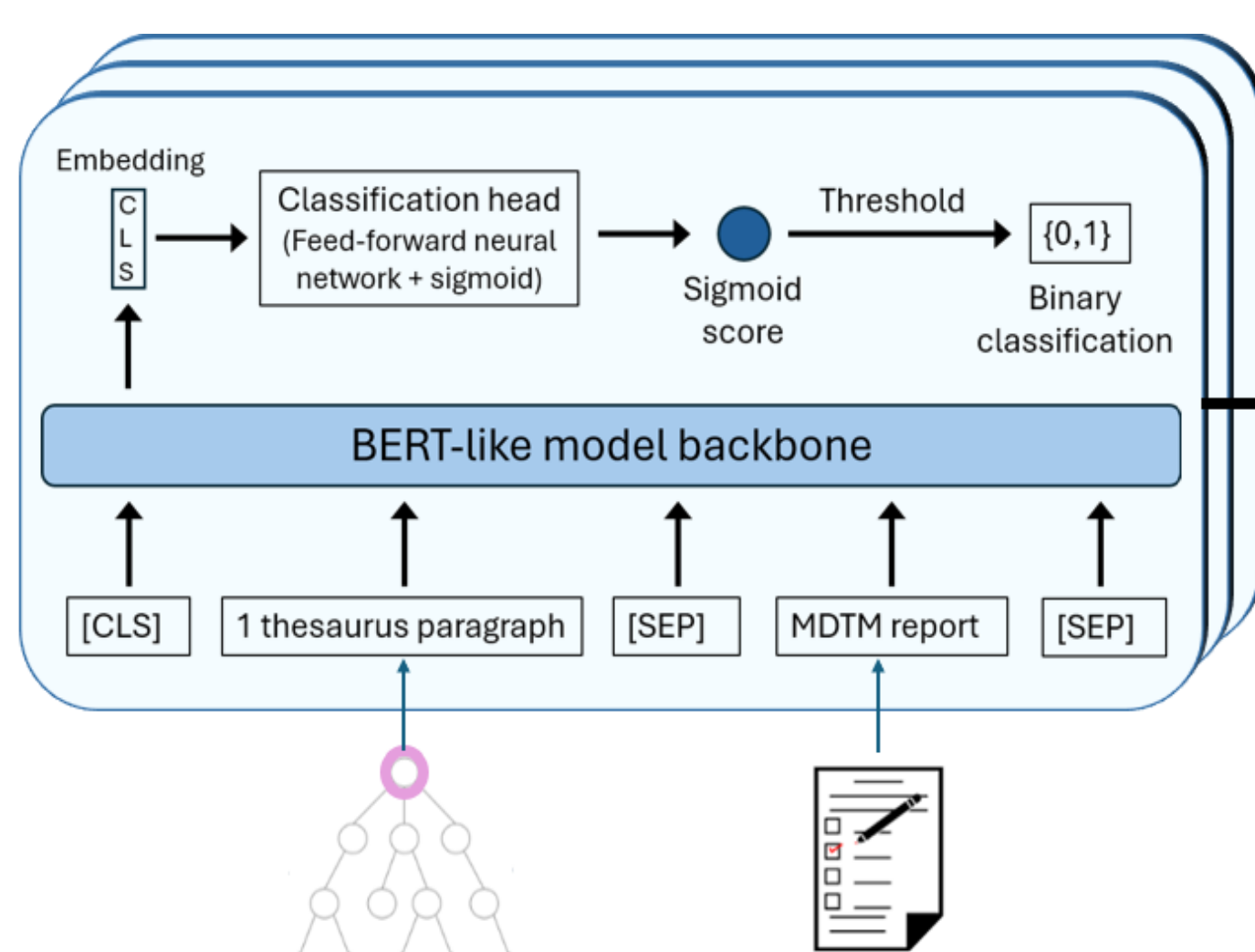
$$F_2 = (1 + 2^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(2^2 \cdot \text{Precision}) + \text{Recall}}$$

- Avoid missing relevant treatments
- Recall is prioritized over precision



Model	Prompt Strategy	F2	Precision	Recall
BioMistral	- basic	0.477	0.165	0.904
	- basic + thesaurus	0.486	0.164	0.952
	- basic + RAG	0.482	0.164	0.934
	- basic + ICL_5	0.530 ± 0.001	0.193 ± 0.000	0.940 ± 0.004
MiXtral	- basic	0.254	0.281	0.247
	- basic + thesaurus	0.459	0.188	0.719
	- basic + RAG	0.465	0.220	0.644
	- basic + ICL_5	0.493 ± 0.008	0.197 ± 0.004	0.793 ± 0.012
Chocolatine	- basic	0.275	0.306	0.268
	- basic + thesaurus	0.397	0.310	0.426
	- basic + RAG	0.435	0.267	0.518
	- basic + ICL_5	0.660 ± 0.012	0.476 ± 0.010	0.730 ± 0.013
Aloe	- basic	0.514	0.213	0.794
	- basic + thesaurus	0.456	0.214	0.636
	- basic + RAG	0.542	0.215	0.875
	- basic + ICL_5	0.570 ± 0.003	0.308 ± 0.005	0.725 ± 0.004
Qwen	- basic	0.241	0.409	0.219
	- basic + thesaurus	0.306	0.346	0.297
	- basic + RAG	0.333	0.309	0.339
	- basic + ICL_5	0.736 ± 0.003	0.678 ± 0.005	0.752 ± 0.003

Our proposed approach: finetuned lightweight crossencoder



Physicians don't read MDTM reports linearly:

1. **Filtering:** Discard fields irrelevant to colorectal cancer.
2. **Prioritization:** Check critical MDTM fields first.
3. **Conditions:** Consult additional fields conditionally based on key field values.

We designed an algorithm that follows their hierarchical clinical reasoning process.

- ❖ **Local lightweight model:** ensures data privacy.
- ❖ **Traceable Recommendations:** Provides source paragraphs directly from the thesaurus

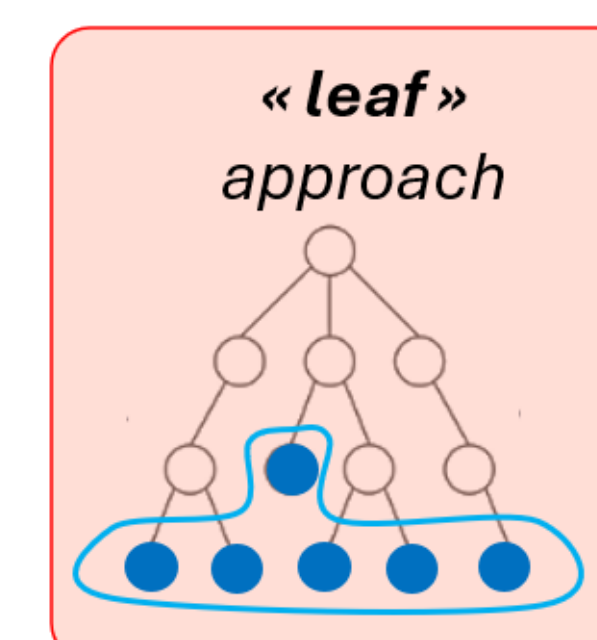
Relevant MDTM fields

Highest-priority fields

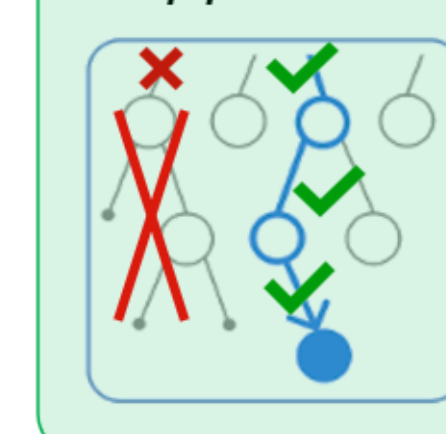
Adding lower-priority fields

If field A = x, then include field B and exclude field C

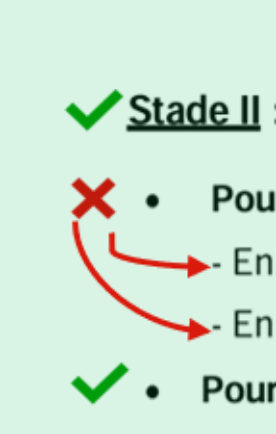
Leaf approach: classifying only leaf paragraphs (treatments are located in the leaves).
Traversal approach: efficiently traversing the thesaurus graph from root to leaves.



« leaf » approach



« traversal » approach



- Stade I :**
- Pas de traitement adjuvant
- Stade II :** (figure 3 et tableau 4)
- Pour les cancers de stade II MSI :
 - ➔ En l'absence de facteurs de mauvais pronostic : pas de chimiothérapie adjuvante
 - ➔ En cas de facteurs de mauvais pronostic : pas de référence
 - Pour les cancers de stade II MSS : pas de chimiothérapie adjuvante

Our approach	Scores on the positive class ± standard deviation			Evaluated node count
	F2	Precision	Recall	
full_MDTM:				
- leaf	0.858 ± 0.024	0.694 ± 0.062	0.914 ± 0.014	70.0
- traversal	0.868 ± 0.036	0.745 ± 0.058	0.907 ± 0.040	36.9
cleaned_MDTM:				
- leaf	0.912 ± 0.013	0.788 ± 0.030	0.950 ± 0.008	70.0
- traversal	0.917 ± 0.021	0.824 ± 0.043	0.944 ± 0.023	34.6
biLevel_MDTM				
- leaf	0.919 ± 0.005	0.796 ± 0.017	0.955 ± 0.005	70.0
- traversal	0.937 ± 0.006	0.869 ± 0.007	0.955 ± 0.007	34.0
branch_MDTM				
- leaf	0.938 ± 0.004	0.827 ± 0.016	0.970 ± 0.003	70.0
- traversal	0.945 ± 0.004	0.866 ± 0.017	0.967 ± 0.007	34.4
perNode_MDTM				
- leaf	0.938 ± 0.005	0.855 ± 0.012	0.961 ± 0.007	70.0
- traversal	0.953 ± 0.005	0.893 ± 0.022	0.969 ± 0.005	34.7